

FIGURE 4.3.20 $E(t)$ in Problem 73

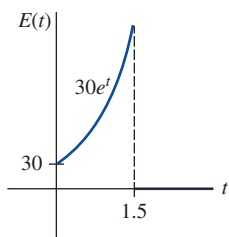


FIGURE 4.3.21 $E(t)$ in Problem 74

75. (a) Use the Laplace transform to find the current $i(t)$ in a single-loop LR -series circuit when $i(0) = 0$, $L = 1$ h, $R = 10 \Omega$, and $E(t)$ is as given in FIGURE 4.3.22.
- (b) Use a computer graphing program to graph $i(t)$ for $0 \leq t \leq 6$. Use the graph to estimate $i_{\max}$ and $i_{\min}$, the maximum and minimum values of the current, respectively.

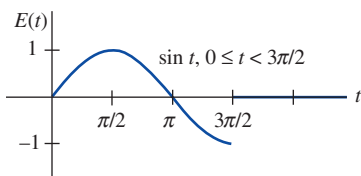


FIGURE 4.3.22 $E(t)$ in Problem 75

76. (a) Use the Laplace transform to find the charge $q(t)$ on the capacitor in an RC -series circuit when $q(0) = 0$, $R = 50 \Omega$, $C = 0.01$ f, and $E(t)$ is as given in FIGURE 4.3.23.
- (b) Assume $E_0 = 100$ V. Use a computer graphing program to graph $q(t)$ for $0 \leq t \leq 6$. Use the graph to estimate $q_{\max}$, the maximum value of the charge.

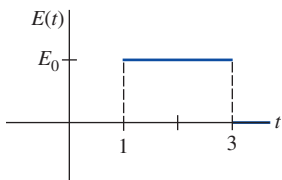


FIGURE 4.3.23 $E(t)$ in Problem 76

77. A cantilever beam is embedded at its left end and free at its right end. Use the Laplace transform to find the deflection $y(x)$ when the load is given by

$$w(x) = \begin{cases} w_0, & 0 < x < L/2 \\ 0, & L/2 \leq x < L. \end{cases}$$

78. Solve Problem 77 when the load is given by

$$w(x) = \begin{cases} 0, & 0 < x < L/3 \\ w_0, & L/3 \leq x < 2L/3 \\ 0, & 2L/3 \leq x < L. \end{cases}$$

79. Find the deflection $y(x)$ of a cantilever beam embedded at its left end and free at its right end when the load is as given in Example 10.
80. A beam is embedded at its left end and simply supported at its right end. Find the deflection $y(x)$ when the load is as given in Problem 77.
81. **Cake Inside an Oven** Reread Example 4 in Section 2.7 on the cooling of a cake that is taken out of an oven.
- (a) Devise a mathematical model for the temperature of a cake while it is *inside* the oven based on the following assumptions: At $t = 0$ the cake mixture is at the room temperature of 70° ; the oven is not preheated so that at $t = 0$, when the cake mixture is placed into the oven, the temperature inside the oven is also 70° ; the temperature of the oven increases linearly until $t = 4$ minutes, when the desired temperature of 300° is attained; the oven temperature is a constant 300° for $t \geq 4$.
- (b) Use the Laplace transform to solve the initial-value problem in part (a).

Discussion Problems

82. Discuss how you would fix up each of the following functions so that Theorem 4.3.2 could be used directly to find the given Laplace transform. Check your answers using (16) of this section.
- (a) $\mathcal{L}\{(2t + 1)\mathcal{U}(t - 1)\}$
- (b) $\mathcal{L}\{e^t\mathcal{U}(t - 5)\}$
- (c) $\mathcal{L}\{\cos t\mathcal{U}(t - \pi)\}$
- (d) $\mathcal{L}\{(t^2 - 3t)\mathcal{U}(t - 2)\}$
83. (a) Assume that Theorem 4.3.1 holds when the symbol a is replaced by ki , where k is a real number and $i^2 = -1$. Show that $\mathcal{L}\{te^{kti}\}$ can be used to deduce

$$\mathcal{L}\{t \cos kt\} = \frac{s^2 - k^2}{(s^2 + k^2)^2}$$

$$\text{and} \quad \mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}.$$

- (b) Now use the Laplace transform to solve the initial-value problem

$$x'' + \omega^2 x = \cos \omega t, \quad x(0) = 0, \quad x'(0) = 0.$$

4.4

Additional Operational Properties

INTRODUCTION In this section we develop several more operational properties of the Laplace transform. Specifically, we shall see how to find the transform of a function $f(t)$ that is multiplied by a monomial t^n , the transform of a special type of integral, and the transform of a periodic function. The last two transform properties allow us to solve some equations that we have not encountered up to this point: Volterra integral equations, integrodifferential equations, and ordinary differential equations in which the input function is a periodic piecewise-defined function.

4.4.1 Derivatives of Transforms

■ Multiplying a Function by t^n The Laplace transform of the product of a function $f(t)$ with t can be found by differentiating the Laplace transform of $f(t)$. If $F(s) = \mathcal{L}\{f(t)\}$ and if we assume that interchanging of differentiation and integration is possible, then

$$\frac{d}{ds} F(s) = \frac{d}{ds} \int_0^\infty e^{-st} f(t) dt = \int_0^\infty \frac{\partial}{\partial s} [e^{-st} f(t)] dt = - \int_0^\infty e^{-st} t f(t) dt = -\mathcal{L}\{t f(t)\};$$

that is,
$$\mathcal{L}\{t f(t)\} = -\frac{d}{ds} \mathcal{L}\{f(t)\}.$$

Similarly,
$$\begin{aligned} \mathcal{L}\{t^2 f(t)\} &= \mathcal{L}\{t \cdot t f(t)\} = -\frac{d}{ds} \mathcal{L}\{t f(t)\} \\ &= -\frac{d}{ds} \left(-\frac{d}{ds} \mathcal{L}\{f(t)\} \right) = \frac{d^2}{ds^2} \mathcal{L}\{f(t)\}. \end{aligned}$$

The preceding two cases suggest the following general result for $\mathcal{L}\{t^n f(t)\}$.

Theorem 4.4.1 Derivatives of Transforms

If $F(s) = \mathcal{L}\{f(t)\}$ and $n = 1, 2, 3, \dots$, then

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s).$$

EXAMPLE 1 Using Theorem 4.4.1

Evaluate $\mathcal{L}\{t \sin kt\}$.

SOLUTION With $f(t) = \sin kt$, $F(s) = k/(s^2 + k^2)$, and $n = 1$, Theorem 4.4.1 gives

$$\mathcal{L}\{t \sin kt\} = -\frac{d}{ds} \mathcal{L}\{\sin kt\} = -\frac{d}{ds} \left(\frac{k}{s^2 + k^2} \right) = \frac{2ks}{(s^2 + k^2)^2}. \quad \equiv$$

If we wanted to evaluate $\mathcal{L}\{t^2 \sin kt\}$ and $\mathcal{L}\{t^3 \sin kt\}$, all we need do, in turn, is take the negative of the derivative with respect to s of the result in Example 1 and then take the negative of the derivative with respect to s of $\mathcal{L}\{t^2 \sin kt\}$.

To find transforms of functions $t^n e^{at}$ we can use either the first translation theorem or Theorem 4.4.1. For example,

$$\text{Theorem 4.3.1: } \mathcal{L}\{te^{3t}\} = \mathcal{L}\{t\}_{s \rightarrow s-3} = \frac{1}{s^2} \Big|_{s \rightarrow s-3} = \frac{1}{(s-3)^2}$$

$$\text{Theorem 4.4.1: } \mathcal{L}\{te^{3t}\} = -\frac{d}{ds} \mathcal{L}\{e^{3t}\} = -\frac{d}{ds} \frac{1}{s-3} = (s-3)^{-2} = \frac{1}{(s-3)^2}.$$

Note that either theorem could be used. 

EXAMPLE 2 An Initial-Value Problem

Solve $x'' + 16x = \cos 4t$, $x(0) = 0$, $x'(0) = 1$.

SOLUTION The initial-value problem could describe the forced, undamped, and resonant motion of a mass on a spring. The mass starts with an initial velocity of 1 foot per second in the downward direction from the equilibrium position.

Transforming the differential equation gives

$$(s^2 + 16)X(s) = 1 + \frac{s}{s^2 + 16} \quad \text{or} \quad X(s) = \frac{1}{s^2 + 16} + \frac{s}{(s^2 + 16)^2}.$$

Now we have just learned in Example 1 that

$$\mathcal{L}^{-1}\left\{\frac{2ks}{(s^2 + k^2)^2}\right\} = t \sin kt, \quad (1)$$

and so with the identification $k = 4$ in (1) and in part (d) of Theorem 4.2.1, we obtain

$$\begin{aligned} x(t) &= \frac{1}{4} \mathcal{L}^{-1}\left\{\frac{4}{s^2 + 16}\right\} + \frac{1}{8} \mathcal{L}^{-1}\left\{\frac{8s}{(s^2 + 16)^2}\right\} \\ &= \frac{1}{4} \sin 4t + \frac{1}{8} t \sin 4t. \end{aligned} \quad \equiv$$

4.4.2 Transforms of Integrals

Convolution If functions f and g are piecewise continuous on the interval $[0, \infty)$, then the **convolution** of f and g , denoted by the symbol $f * g$, is a function defined by the integral

$$f * g = \int_0^t f(\tau)g(t - \tau) d\tau. \quad (2)$$

Because we are integrating in (2) with respect to the variable τ (the lowercase Greek letter *tau*), the convolution $f * g$ is a function of t . To emphasize this fact, (2) is also written $(f * g)(t)$. As the notation $f * g$ suggests, the convolution (2) is often interpreted as a *generalized product* of two functions f and g .

EXAMPLE 3 Convolution of Two Functions

Evaluate (a) $e^t * \sin t$ (b) $\mathcal{L}\{e^t * \sin t\}$.

SOLUTION (a) With the identifications

$$f(t) = e^t, g(t) = \sin t \quad \text{and} \quad f(\tau) = e^\tau, g(t - \tau) = \sin(t - \tau)$$

it follows from (2) and integration by parts that

$$\begin{aligned} e^t * \sin t &= \int_0^t e^\tau \sin(t - \tau) d\tau \\ &= \frac{1}{2} [e^\tau \sin(t - \tau) + e^\tau \cos(t - \tau)]_0^t \\ &= \frac{1}{2} (-\sin t - \cos t + e^t). \end{aligned} \quad (3)$$

(b) Then from (3) and parts (c), (d), and (e) of Theorem 4.1.1 we find

$$\begin{aligned} \mathcal{L}\{e^t * \sin t\} &= -\frac{1}{2} \mathcal{L}\{\sin t\} - \frac{1}{2} \mathcal{L}\{\cos t\} + \frac{1}{2} \mathcal{L}\{e^t\} \\ &= -\frac{1}{2} \frac{1}{s^2 + 1} - \frac{1}{2} \frac{s}{s^2 + 1} + \frac{1}{2} \frac{1}{s - 1} \\ &= \frac{1}{(s - 1)(s^2 + 1)}. \end{aligned} \quad \equiv$$

It is left as an exercise to show that

$$\int_0^t f(\tau)g(t - \tau) d\tau = \int_0^t f(t - \tau)g(\tau) d\tau,$$

that is, $f * g = g * f$. In other words, the convolution of two functions is commutative.

Convolution Theorem We have seen that if f and g are both piecewise continuous for $t \geq 0$, then the Laplace transform of a sum $f + g$ is the sum of the individual Laplace transforms. While it is *not* true that the Laplace transform of the product fg is the product of the Laplace

transforms, we see in the next theorem—called the **convolution theorem**—that the Laplace transform of the generalized product $f * g$ is the product of the Laplace transforms of f and g .

Theorem 4.4.2 Convolution Theorem

If $f(t)$ and $g(t)$ are piecewise continuous on $[0, \infty)$ and of exponential order, then

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\} = F(s)G(s).$$

PROOF: Let

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-s\tau} f(\tau) d\tau \quad \text{and} \quad G(s) = \mathcal{L}\{g(t)\} = \int_0^\infty e^{-s\beta} g(\beta) d\beta.$$

Proceeding formally, we have

$$\begin{aligned} F(s)G(s) &= \left(\int_0^\infty e^{-s\tau} f(\tau) d\tau \right) \left(\int_0^\infty e^{-s\beta} g(\beta) d\beta \right) \\ &= \int_0^\infty \int_0^\infty e^{-s(\tau+\beta)} f(\tau)g(\beta) d\tau d\beta \\ &= \int_0^\infty f(\tau) d\tau \int_0^\infty e^{-s(\tau+\beta)} g(\beta) d\beta. \end{aligned}$$

Holding τ fixed, we let $t = \tau + \beta$, $dt = d\beta$, so that

$$F(s)G(s) = \int_0^\infty f(\tau) d\tau \int_\tau^\infty e^{-st} g(t - \tau) dt.$$

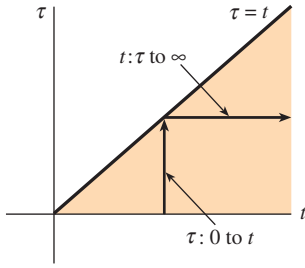


FIGURE 4.4.1 Changing order of integration from t first to τ first

In the $t\tau$ -plane we are integrating over the shaded region in **FIGURE 4.4.1**. Since f and g are piecewise continuous on $[0, \infty)$ and of exponential order, it is possible to interchange the order of integration:

$$F(s)G(s) = \int_0^\infty e^{-st} dt \int_0^t f(\tau)g(t - \tau) d\tau = \int_0^\infty e^{-st} \left\{ \int_0^t f(\tau)g(t - \tau) d\tau \right\} dt = \mathcal{L}\{f * g\}. \quad \equiv$$

Theorem 4.4.2 shows that we can find the Laplace transform of the convolution $f * g$ of two functions without actually evaluating the definite integral $\int_0^t f(\tau)g(t - \tau) d\tau$ as we did in (3). The next example illustrates the idea.

EXAMPLE 4 Using Theorem 4.4.2

Evaluate $\mathcal{L}\left\{ \int_0^t e^\tau \sin(t - \tau) d\tau \right\}$.

SOLUTION This is the same as the transform $\mathcal{L}\{e^t * \sin t\}$ that we found part (b) of Example 3. This time we use Theorem 4.4.2 that the Laplace transform of the convolution of f and g is the product of their Laplace transforms:

$$\begin{aligned} \mathcal{L}\left\{ \int_0^t e^\tau \sin(t - \tau) d\tau \right\} &= \mathcal{L}\{e^t * \sin t\} \\ &= \mathcal{L}\{e^t\} \cdot \mathcal{L}\{\sin t\} \\ &= \frac{1}{s - 1} \cdot \frac{1}{s^2 + 1} \\ &= \frac{1}{(s - 1)(s^2 + 1)}. \quad \equiv \end{aligned}$$

Inverse Form of Theorem 4.4.2 The convolution theorem is sometimes useful in finding the inverse Laplace transforms of the product of two Laplace transforms. From Theorem 4.4.2 we have

$$\mathcal{L}^{-1}\{F(s)G(s)\} = f * g. \quad (4)$$

Many of the results in the table of Laplace transforms in Appendix III can be derived using (4). For instance, in the next example we obtain entry 25 of the table:

$$\mathcal{L}\{\sin kt - kt \cos kt\} = \frac{2k^3}{(s^2 + k^2)^2}. \quad (5)$$

EXAMPLE 5 Inverse Transform as a Convolution

Evaluate $\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + k^2)^2}\right\}$.

SOLUTION Let $F(s) = G(s) = \frac{1}{s^2 + k^2}$

so that $f(t) = g(t) = \frac{1}{k} \mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\} = \frac{1}{k} \sin kt$.

In this case (4) gives

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + k^2)^2}\right\} = \frac{1}{k^2} \int_0^t \sin k\tau \sin k(t - \tau) d\tau. \quad (6)$$

Now recall from trigonometry that

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)].$$

If we set $A = k\tau$ and $B = k(t - \tau)$, we can carry out the integration in (6):

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{1}{(s^2 + k^2)^2}\right\} &= \frac{1}{2k^2} \int_0^t [\cos k(2\tau - t) - \cos kt] d\tau \\ &= \frac{1}{2k^2} \left[\frac{1}{2k} \sin k(2\tau - t) - \tau \cos kt \right]_0^t \\ &= \frac{\sin kt - kt \cos kt}{2k^3}. \end{aligned}$$

Multiplying both sides by $2k^3$ gives the inverse form of (5). ≡

Transform of an Integral When $g(t) = 1$ and $\mathcal{L}\{g(t)\} = G(s) = 1/s$, the convolution theorem implies that the Laplace transform of the integral of f is

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}. \quad (7)$$

The inverse form of (7),

$$\int_0^t f(\tau) d\tau = \mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\}, \quad (8)$$

can be used in lieu of partial fractions when s^n is a factor of the denominator and $f(t) = \mathcal{L}^{-1}\{F(s)\}$ is easy to integrate. For example, we know for $f(t) = \sin t$ that $F(s) = 1/(s^2 + 1)$, and so by (8)

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/(s^2 + 1)}{s}\right\} = \int_0^t \sin \tau \, d\tau = 1 - \cos t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2 + 1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s(s^2 + 1)}{s}\right\} = \int_0^t (1 - \cos \tau) \, d\tau = t - \sin t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3(s^2 + 1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s^2(s^2 + 1)}{s}\right\} = \int_0^t (\tau - \sin \tau) \, d\tau = \frac{1}{2}t^2 - 1 + \cos t$$

and so on.

Volterra Integral Equation The convolution theorem and the result in (7) are useful in solving other types of equations in which an unknown function appears under an integral sign. In the next example, we solve a **Volterra integral equation** for $f(t)$,

$$f(t) = g(t) + \int_0^t f(\tau) h(t - \tau) \, d\tau. \quad (9)$$

The functions $g(t)$ and $h(t)$ are known. Notice that the integral in (9) has the convolution form (2) with the symbol h playing the part of g .

EXAMPLE 6 An Integral Equation

Solve $f(t) = 3t^2 - e^{-t} - \int_0^t f(\tau) e^{t-\tau} \, d\tau$ for $f(t)$.

SOLUTION In the integral we identify $h(t - \tau) = e^{t-\tau}$ so that $h(t) = e^t$. We take the Laplace transform of each term; in particular, by Theorem 4.4.2 the transform of the integral is the product of $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{e^t\} = 1/(s - 1)$:

$$F(s) = 3 \cdot \frac{2}{s^3} - \frac{1}{s + 1} - F(s) \cdot \frac{1}{s - 1}.$$

After solving the last equation for $F(s)$ and carrying out the partial fraction decomposition, we find

$$F(s) = \frac{6}{s^3} - \frac{6}{s^4} + \frac{1}{s} - \frac{2}{s + 1}.$$

The inverse transform then gives

$$\begin{aligned} f(t) &= 3\mathcal{L}^{-1}\left\{\frac{2!}{s^3}\right\} - \mathcal{L}^{-1}\left\{\frac{3!}{s^4}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{s + 1}\right\} \\ &= 3t^2 - t^3 + 1 - 2e^{-t}. \end{aligned}$$

Series Circuits In a single-loop or series circuit, Kirchhoff's second law states that the sum of the voltage drops across an inductor, resistor, and capacitor is equal to the impressed voltage $E(t)$. Now it is known that the voltage drops across an inductor, resistor, and capacitor are, respectively,

$$L \frac{di}{dt}, \quad R i(t), \quad \text{and} \quad \frac{1}{C} \int_0^t i(\tau) \, d\tau,$$

where $i(t)$ is the current and L , R , and C are constants. It follows that the current in an **LRC-series circuit**, such as that shown in **FIGURE 4.4.2**, is governed by the **integrodifferential equation**

$$L \frac{di}{dt} + R i(t) + \frac{1}{C} \int_0^t i(\tau) \, d\tau = E(t). \quad (10)$$

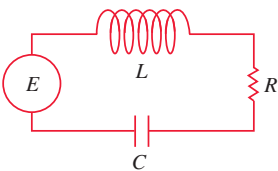


FIGURE 4.4.2 LRC-series circuit

EXAMPLE 7**An Integrodifferential Equation**

Determine the current $i(t)$ in a single-loop LRC -circuit when $L = 0.1$ h, $R = 2 \Omega$, $C = 0.1$ f, $i(0) = 0$, and the impressed voltage is

$$E(t) = 120t - 120t\mathcal{U}(t - 1).$$

SOLUTION Using the given data, equation (10) becomes

$$0.1 \frac{di}{dt} + 2i + 10 \int_0^t i(\tau) d\tau = 120t - 120t\mathcal{U}(t - 1).$$

Now by (7), $\mathcal{L}\{\int_0^t i(\tau) d\tau\} = I(s)/s$, where $I(s) = \mathcal{L}\{i(t)\}$. Thus the Laplace transform of the integrodifferential equation is

$$0.1sI(s) + 2I(s) + 10 \frac{I(s)}{s} = 120 \left[\frac{1}{s^2} - \frac{1}{s^2}e^{-s} - \frac{1}{s}e^{-s} \right]. \leftarrow \text{by (16) of Section 4.3}$$

Multiplying this equation by $10s$, using $s^2 + 20s + 100 = (s + 10)^2$, and then solving for $I(s)$ gives

$$I(s) = 1200 \left[\frac{1}{s(s + 10)^2} - \frac{1}{s(s + 10)^2}e^{-s} - \frac{1}{(s + 10)^2}e^{-s} \right].$$

By partial fractions,

$$\begin{aligned} I(s) = 1200 & \left[\frac{1/100}{s} - \frac{1/100}{s + 10} - \frac{1/10}{(s + 10)^2} - \frac{1/100}{s}e^{-s} \right. \\ & \left. + \frac{1/100}{s + 10}e^{-s} + \frac{1/10}{(s + 10)^2}e^{-s} - \frac{1}{(s + 10)^2}e^{-s} \right]. \end{aligned}$$

From the inverse form of the second translation theorem, (15) of Section 4.3, we finally obtain

$$\begin{aligned} i(t) = 12[1 - \mathcal{U}(t - 1)] - 12[e^{-10t} - e^{-10(t-1)}\mathcal{U}(t - 1)] \\ - 120te^{-10t} - 1080(t - 1)e^{-10(t-1)}\mathcal{U}(t - 1). \end{aligned}$$

Written as a piecewise-defined function, the current is

$$i(t) = \begin{cases} 12 - 12e^{-10t} - 120te^{-10t}, & 0 \leq t < 1 \\ -12e^{-10t} + 12e^{-10(t-1)} - 120te^{-10t} - 1080(t - 1)e^{-10(t-1)}, & t \geq 1. \end{cases} \quad (11)$$

Using the last form of the solution and a CAS, we graph $i(t)$ on each of the two intervals and then combine the graphs. Note in **FIGURE 4.4.3** that even though the input $E(t)$ is discontinuous, the output or response $i(t)$ is a continuous function. $\equiv$

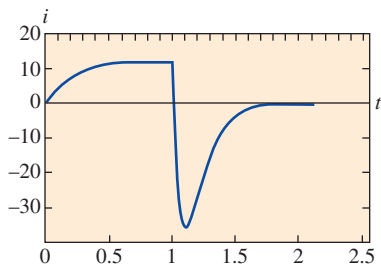


FIGURE 4.4.3 Graph of current $i(t)$ in (11) of Example 7

Post Script—Green's Function Redux By applying the Laplace transform to the initial-value problem

$$y'' + ay' + by = f(t), \quad y(0) = 0, y'(0) = 0,$$

where a and b are constants, we find that the transform of $y(t)$ is

$$Y(s) = \frac{F(s)}{s^2 + as + b},$$

where $F(s) = \mathcal{L}\{f(t)\}$. By rewriting the foregoing transform as the product

$$Y(s) = \frac{1}{s^2 + as + b} F(s)$$

we can use the inverse form of the convolution theorem (4) to write the solution of the IVP as

$$y(t) = \int_0^t g(t - \tau) f(\tau) d\tau, \quad (12)$$

where $\mathcal{L}^{-1}\left\{\frac{1}{s^2 + as + b}\right\} = g(t)$ and $\mathcal{L}^{-1}\{F(s)\} = f(t)$. On the other hand, we know from (9) of Section 3.10 that the solution of the IVP is also given by

$$y(t) = \int_0^t G(t, \tau) f(\tau) d\tau, \quad (13)$$

where $G(t, \tau)$ is the Green's function for the differential equation.

By comparing (12) and (13) we see that the Green's function for the differential equation is related to $\mathcal{L}^{-1}\left\{\frac{1}{s^2 + as + b}\right\} = g(t)$ by

$$G(t, \tau) = g(t - \tau). \quad (14)$$

For example, for the initial-value problem $y'' + 4y = f(t)$, $y(0) = 0$, $y'(0) = 0$ we find

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4}\right\} = \frac{1}{2} \sin 2t = g(t).$$

In Example 4 of Section 3.10, the roles of the symbols x and t are played by t and τ , respectively, in this discussion.

Thus from (14) we see that the Green's function for the DE $y'' + 4y = f(t)$ is $G(t, \tau) = g(t - \tau) = \frac{1}{2} \sin 2(t - \tau)$. See Example 4 in Section 3.10.

REMARKS

Although the defining integral in the Laplace transform was known and used for a long time prior to the twentieth century, it was not used to solve differential equations. The fact that we use the Laplace transform today to solve a variety of equations is actually due to Oliver Heaviside (see page 229). In 1893, Heaviside invented an operational calculus for solving differential equations encountered in electrical engineering. Heaviside was no mathematician, and his procedures for solving differential equations were formal manipulations or procedures lacking mathematical justification. Nonetheless, these procedures worked. In an attempt to put his operational calculus on a sound foundation, mathematicians discovered that the rules of his calculus matched many properties of the Laplace transform. Over time, Heaviside's operation calculus disappeared, to be replaced by the theory and applications of the Laplace transform.

You should verify either by substitution in the equation or by the method of Section 2.3 that $y(t) = e^{-t} \int_0^t e^{u^2+u} du$ is a perfectly good solution of the linear initial-value problem $y' + y = e^{t^2}$, $y(0) = 0$. We now solve the same equation with a formal application of the Laplace transform. If we denote $\mathcal{L}\{y\} = Y(s)$ and $\mathcal{L}\{e^{t^2}\} = F(s)$, then the transform of the equation is

$$sY(s) - y(0) + Y(s) = F(s) \quad \text{or} \quad Y(s) = \frac{F(s)}{s + 1}.$$

Using $\mathcal{L}^{-1}\{F(s)\} = e^{t^2}$ and $\mathcal{L}^{-1}\left\{\frac{1}{s + 1}\right\} = e^{-t}$ it follows from the inverse form (4) of the convolution theorem that the solution of the initial-value problem is

$$y(t) = \mathcal{L}^{-1}\left\{F(s) \cdot \frac{1}{s + 1}\right\} = \int_0^t e^{\tau^2} \cdot e^{-(t-\tau)} d\tau = e^{-t} \int_0^t e^{\tau^2+\tau} d\tau.$$

With τ playing the part of u , this is the solution as first given. Do you see anything strange in this solution method of the IVP?

4.4.3 Transform of a Periodic Function

Periodic Function If a periodic function f has period T , $T > 0$, then $f(t + T) = f(t)$. The Laplace transform of a periodic function can be obtained by integration over one period.

Theorem 4.4.3 Transform of a Periodic Function

If $f(t)$ is piecewise continuous on $[0, \infty)$, of exponential order, and periodic with period T , then

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

PROOF: Write the Laplace transform of f as two integrals:

$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t) dt + \int_T^\infty e^{-st} f(t) dt.$$

When we let $t = u + T$, the last integral becomes

$$\int_T^\infty e^{-st} f(t) dt = \int_0^\infty e^{-s(u+T)} f(u+T) du = e^{-sT} \int_0^\infty e^{-su} f(u) du = e^{-sT} \mathcal{L}\{f(t)\}.$$

Therefore
$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t) dt + e^{-sT} \mathcal{L}\{f(t)\}.$$

Solving the equation in the last line for $\mathcal{L}\{f(t)\}$ proves the theorem. $\equiv$

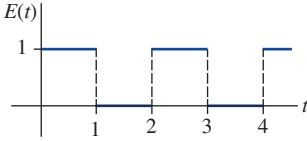


FIGURE 4.4.4 Square wave in Example 8

EXAMPLE 8 Transform of a Periodic Function

Find the Laplace transform of the periodic function shown in FIGURE 4.4.4.

SOLUTION The function $E(t)$ is called a square wave and has period $T = 2$. For $0 \leq t < 2$, $E(t)$ can be defined by

$$E(t) = \begin{cases} 1, & 0 \leq t < 1 \\ 0, & 1 \leq t < 2, \end{cases}$$

and outside the interval by $f(t + 2) = f(t)$. Now from Theorem 4.4.3,

$$\begin{aligned} \mathcal{L}\{E(t)\} &= \frac{1}{1 - e^{-2s}} \int_0^2 e^{-st} E(t) dt = \frac{1}{1 - e^{-2s}} \left[\int_0^1 e^{-st} \cdot 1 dt + \int_1^2 e^{-st} \cdot 0 dt \right] \\ &= \frac{1}{1 - e^{-2s}} \frac{1 - e^{-s}}{s} \quad \leftarrow 1 - e^{-2s} = (1 + e^{-s})(1 - e^{-s}) \\ &= \frac{1}{s(1 + e^{-s})}. \end{aligned} \quad (15) \quad \equiv$$

EXAMPLE 9 A Periodic Impressed Voltage

The differential equation for the current $i(t)$ in a single-loop LR -series circuit is

$$L \frac{di}{dt} + Ri = E(t). \quad (16)$$

Determine the current $i(t)$ when $i(0) = 0$ and $E(t)$ is the square-wave function given in Figure 4.4.4.

SOLUTION Using the result in (15) of the preceding example, the Laplace transform of the DE is

$$LsI(s) + RI(s) = \frac{1}{s(1 + e^{-s})} \quad \text{or} \quad I(s) = \frac{1/L}{s(s + R/L)} \cdot \frac{1}{1 + e^{-s}}. \quad (17)$$

To find the inverse Laplace transform of the last function, we first make use of geometric series. With the identification $x = e^{-s}$, $s > 0$, the geometric series

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots \quad \text{becomes} \quad \frac{1}{1+e^{-s}} = 1 - e^{-s} + e^{-2s} - e^{-3s} + \cdots.$$

From
$$\frac{1}{s(s+R/L)} = \frac{L/R}{s} - \frac{L/R}{s+R/L},$$

we can then rewrite (17) as

$$\begin{aligned} I(s) &= \frac{1}{R} \left(\frac{1}{s} - \frac{1}{s+R/L} \right) (1 - e^{-s} + e^{-2s} - e^{-3s} + \cdots) \\ &= \frac{1}{R} \left(\frac{1}{s} - \frac{e^{-s}}{s} + \frac{e^{-2s}}{s} - \frac{e^{-3s}}{s} + \cdots \right) - \frac{1}{R} \left(\frac{1}{s+R/L} - \frac{e^{-s}}{s+R/L} + \frac{e^{-2s}}{s+R/L} - \frac{e^{-3s}}{s+R/L} + \cdots \right). \end{aligned}$$

By applying the form of the second translation theorem to each term of both series we obtain

$$\begin{aligned} i(t) &= \frac{1}{R} \left(1 - \mathcal{U}(t-1) + \mathcal{U}(t-2) - \mathcal{U}(t-3) + \cdots \right) \\ &\quad - \frac{1}{R} \left(e^{-Rt/L} - e^{-R(t-1)/L} \mathcal{U}(t-1) + e^{-R(t-2)/L} \mathcal{U}(t-2) - e^{-R(t-3)/L} \mathcal{U}(t-3) + \cdots \right) \end{aligned}$$

or, equivalently,

$$i(t) = \frac{1}{R} (1 - e^{-Rt/L}) + \frac{1}{R} \sum_{n=1}^{\infty} (-1)^n (1 - e^{-R(t-n)/L}) \mathcal{U}(t-n).$$

To interpret the solution, let us assume for the sake of illustration that $R = 1$, $L = 1$, and $0 \leq t < 4$. In this case

$$i(t) = 1 - e^{-t} - (1 - e^{-(t-1)}) \mathcal{U}(t-1) + (1 - e^{-(t-2)}) \mathcal{U}(t-2) - (1 - e^{-(t-3)}) \mathcal{U}(t-3);$$

in other words,

$$i(t) = \begin{cases} 1 - e^{-t}, & 0 \leq t < 1 \\ -e^{-t} + e^{-(t-1)}, & 1 \leq t < 2 \\ 1 - e^{-t} + e^{-(t-1)} - e^{-(t-2)}, & 2 \leq t < 3 \\ -e^{-t} + e^{-(t-1)} - e^{-(t-2)} + e^{-(t-3)}, & 3 \leq t < 4. \end{cases} \quad (18)$$

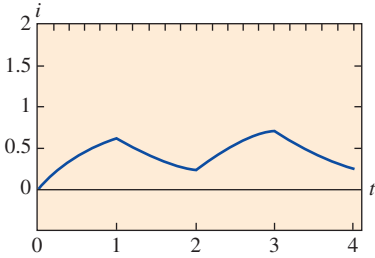


FIGURE 4.4.5 Graph of current $i(t)$ in Example 9

The graph of $i(t)$ for $0 \leq t < 4$, given in **FIGURE 4.4.5**, was obtained with the help of a CAS. $\equiv$

4.4

Exercises

Answers to selected odd-numbered problems begin on page ANS-9.

4.4.1 Derivatives of Transforms

In Problems 1–8, use Theorem 4.4.1 to evaluate the given Laplace transform.

1. $\mathcal{L}\{te^{-10t}\}$
2. $\mathcal{L}\{t^3 e^t\}$
3. $\mathcal{L}\{t \cos 2t\}$
4. $\mathcal{L}\{t \sinh 3t\}$
5. $\mathcal{L}\{t \sinh t\}$
6. $\mathcal{L}\{t^2 \cos t\}$
7. $\mathcal{L}\{te^{2t} \sin 6t\}$
8. $\mathcal{L}\{te^{-3t} \cos 3t\}$

In Problems 9–14, use the Laplace transform to solve the given initial-value problem. Use the table of Laplace transforms in Appendix III as needed.

9. $y' + y = t \sin t$, $y(0) = 0$
10. $y' - y = te^t \sin t$, $y(0) = 0$
11. $y'' + 9y = \cos 3t$, $y(0) = 2$, $y'(0) = 5$
12. $y'' + y = \sin t$, $y(0) = 1$, $y'(0) = -1$

13. $y'' + 16y = f(t)$, $y(0) = 0$, $y'(0) = 1$, where

$$f(t) = \begin{cases} \cos 4t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

14. $y'' + y = f(t)$, $y(0) = 1$, $y'(0) = 0$, where

$$f(t) = \begin{cases} 1, & 0 \leq t < \pi/2 \\ \sin t, & t \geq \pi/2 \end{cases}$$

In Problems 15 and 16, use a graphing utility to graph the indicated solution.

15. $y(t)$ of Problem 13 for $0 \leq t < 2\pi$

16. $y(t)$ of Problem 14 for $0 \leq t < 3\pi$

In some instances the Laplace transform can be used to solve linear differential equations with variable monomial coefficients. In Problems 17 and 18, use Theorem 4.4.1 to reduce the given differential equation to a linear first-order DE in the transformed function $Y(s) = \mathcal{L}\{y(t)\}$. Solve the first-order DE for $Y(s)$ and then find $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.

17. $ty'' - y' = 2t^2$, $y(0) = 0$

18. $2y'' + ty' - 2y = 10$, $y(0) = y'(0) = 0$

4.4.2 Transforms of Integrals

In Problems 19–22, proceed as in Example 3 and find the convolution $f * g$ of the given functions. After integrating find the Laplace transform of $f * g$.

19. $f(t) = 4t$, $g(t) = 3t^2$

20. $f(t) = t$, $g(t) = e^{-t}$

21. $f(t) = e^{-t}$, $g(t) = e^t$

22. $f(t) = \cos 2t$, $g(t) = e^t$

In Problems 23–34, proceed as in Example 4 and find the Laplace transform of $f * g$ using Theorem 4.4.2. Do not evaluate the convolution integral before transforming.

23. $\mathcal{L}\{1 * t^3\}$

24. $\mathcal{L}\{t^2 * te^t\}$

25. $\mathcal{L}\{e^{-t} * e^t \cos t\}$

26. $\mathcal{L}\{e^{2t} * \sin t\}$

27. $\mathcal{L}\left\{\int_0^t e^\tau d\tau\right\}$

28. $\mathcal{L}\left\{\int_0^t \cos \tau d\tau\right\}$

29. $\mathcal{L}\left\{\int_0^t e^{-\tau} \cos \tau d\tau\right\}$

30. $\mathcal{L}\left\{\int_0^t \tau \sin \tau d\tau\right\}$

31. $\mathcal{L}\left\{\int_0^t \tau e^{t-\tau} d\tau\right\}$

32. $\mathcal{L}\left\{\int_0^t \sin \tau \cos(t - \tau) d\tau\right\}$

33. $\mathcal{L}\left\{t \int_0^t \sin \tau d\tau\right\}$

34. $\mathcal{L}\left\{t \int_0^t te^{-\tau} d\tau\right\}$

In Problems 35–38, use (8) to evaluate the given inverse transform.

35. $\mathcal{L}^{-1}\left\{\frac{1}{s(s-1)}\right\}$

36. $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s-1)}\right\}$

37. $\mathcal{L}^{-1}\left\{\frac{1}{s^3(s-1)}\right\}$

38. $\mathcal{L}^{-1}\left\{\frac{1}{s(s-a)^2}\right\}$

39. The table in Appendix III does not contain an entry for

$$\mathcal{L}^{-1}\left\{\frac{8k^3s}{(s^2 + k^2)^3}\right\}.$$

- (a) Use (4) along with the result in (5) to evaluate this inverse transform. Use a CAS as an aid in evaluating the convolution integral.

- (b) Reexamine your answer to part (a). Could you have obtained the result in a different manner?

40. Use the Laplace transform and the result of Problem 39 to solve the initial-value problem

$$y'' + y = \sin t + t \sin t, \quad y(0) = 0, \quad y'(0) = 0.$$

Use a graphing utility to graph the solution.

In Problems 41–50, use the Laplace transform to solve the given integral equation or integrodifferential equation.

41. $f(t) + \int_0^t (t - \tau)f(\tau) d\tau = t$

42. $f(t) = 2t - 4 \int_0^t \sin \tau f(t - \tau) d\tau$

43. $f(t) = te^t + \int_0^t \tau f(t - \tau) d\tau$

44. $f(t) + 2 \int_0^t f(\tau) \cos(t - \tau) d\tau = 4e^{-t} + \sin t$

45. $f(t) + \int_0^t f(\tau) d\tau = 1$

46. $f(t) = \cos t + \int_0^t e^{-\tau} f(t - \tau) d\tau$

47. $f(t) = 1 + t - \frac{8}{3} \int_0^t (\tau - t)^3 f(\tau) d\tau$

48. $t - 2f(t) = \int_0^t (e^\tau - e^{-\tau}) f(t - \tau) d\tau$

49. $y'(t) = 1 - \sin t - \int_0^t y(\tau) d\tau$, $y(0) = 0$

50. $\frac{dy}{dt} + 6y(t) + 9 \int_0^t y(\tau) d\tau = 1$, $y(0) = 0$

In Problems 51 and 52, solve equation (10) subject to $i(0) = 0$ with L , R , C , and $E(t)$ as given. Use a graphing utility to graph the solution for $0 \leq t \leq 3$.

51. $L = 0.1$ h, $R = 3 \Omega$, $C = 0.05$ f,
 $E(t) = 100[\mathcal{U}(t - 1) - \mathcal{U}(t - 2)]$

52. $L = 0.005$ h, $R = 1 \Omega$, $C = 0.02$ f,
 $E(t) = 100[t - (t - 1)\mathcal{U}(t - 1)]$

53. The Laplace transform $\mathcal{L}\{e^{-t^2}\}$ exists, but without finding it solve the initial-value problem

$$y'' + 9y = 3e^{-t^2}, \quad y(0) = 0, \quad y'(0) = 0.$$

54. Solve the integral equation

$$f(t) = e^t + e^t \int_0^t e^{-\tau} f(\tau) d\tau.$$

4.4.3 Transform of a Periodic Function

In Problems 55–60, use Theorem 4.4.3 to find the Laplace transform of the given periodic function.

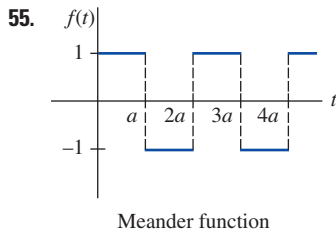


FIGURE 4.4.6 Graph for Problem 55

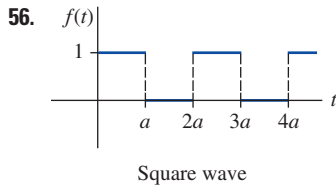


FIGURE 4.4.7 Graph for Problem 56

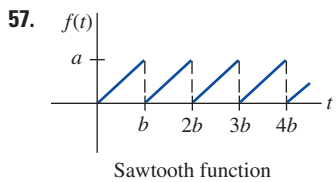


FIGURE 4.4.8 Graph for Problem 57

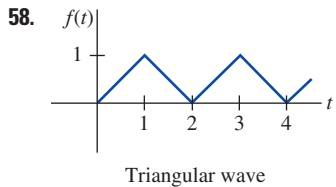


FIGURE 4.4.9 Graph for Problem 58

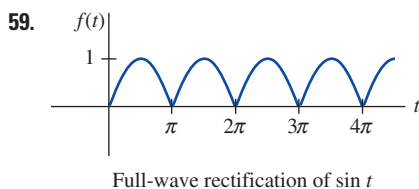


FIGURE 4.4.10 Graph for Problem 59

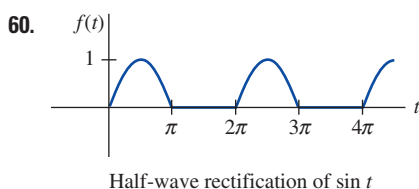


FIGURE 4.4.11 Graph for Problem 60

In Problems 61 and 62, solve equation (16) subject to $i(0) = 0$ with $E(t)$ as given. Use a graphing utility to graph the solution for $0 \leq t \leq 4$ in the case when $L = 1$ and $R = 1$.

61. $E(t)$ is the meander function in Problem 55 with amplitude 1 and $a = 1$.
 62. $E(t)$ is the sawtooth function in Problem 57 with amplitude 1 and $b = 1$.

In Problems 63 and 64, solve the model for a driven spring/mass system with damping

$$m \frac{d^2x}{dt^2} + \beta \frac{dx}{dt} + kx = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

where the driving function f is as specified. Use a graphing utility to graph $x(t)$ for the indicated values of t .

63. $m = \frac{1}{2}$, $\beta = 1$, $k = 5$, f is the meander function in Problem 55 with amplitude 10, and $a = \pi$, $0 \leq t \leq 2\pi$.
 64. $m = 1$, $\beta = 2$, $k = 1$, f is the square wave in Problem 56 with amplitude 5, and $a = \pi$, $0 \leq t \leq 4\pi$.

Discussion Problems

65. Show how to use the Laplace transform to find the numerical value of the improper integral $\int_0^\infty te^{-2t} \sin 4t \, dt$.
 66. In Problem 53 we were able to solve an initial-value problem without knowing the Laplace transform $\mathcal{L}\{e^{-t^2}\}$. In this problem you are asked to find the actual transformed function $Y(s) = \mathcal{L}\{e^{-t^2}\}$ by solving another initial-value problem.
 (a) If $y = e^{-t^2}$, then show that y is a solution of the initial-value problem

$$\frac{dy}{dt} + 2ty = 0, \quad y(0) = 1.$$

- (b) Find $Y(s) = \mathcal{L}\{e^{-t^2}\}$ by using the Laplace transform to solve the problem in part (a). [Hint: First find $Y(0)$ by rereading the material on the error function in Section 2.3. Then in the solution of the resulting linear first-order DE in $Y(s)$ integrate on the interval $[0, s]$. It also helps to use a dummy variable of integration.]

67. Discuss how Theorem 4.4.1 can be used to find

$$\mathcal{L}^{-1}\left\{\ln \frac{s-3}{s+1}\right\}.$$

68. Bessel's differential equation of order $n = 0$ is

$$ty'' + y' + ty = 0.$$

We shall see in Section 5.3 that a solution of the initial-value problem $ty'' + y' + ty = 0$, $y(0) = 1$, $y'(0) = 0$ is $y = J_0(t)$, called the **Bessel function** of the first kind of order $\nu = 0$. Use the procedure outlined in the instructions to Problems 17 and 18 to show that

$$\mathcal{L}\{J_0(t)\} = \frac{1}{\sqrt{s^2 + 1}}.$$

[Hint: You may need to use Problem 52 in Exercises 4.2. Also, it is known that $J_0(0) = 1$.]